

1. Introduction & Objectives

Weather prediction plays an important role in agriculture, transportation, environmental monitoring, and disaster management. With the help of data analysis and machine learning, weather parameters such as temperature, humidity, rainfall, and pollution can be interpreted to identify trends and predict future conditions.

This project analyzes 60 days of synthetic weather data and builds a machine-learning model to predict future temperature values. It also provides a menu-driven interactive system for anomaly detection, visualization, and predictions.

Objectives

- a. To load and preprocess weather data.
 - b. To analyze temperature, rainfall, humidity, and pollution trends.
 - c. To detect anomalies in weather data using z-score.
 - d. To visualize weather patterns through graphs.
 - e. To build a linear regression model for temperature prediction.
 - f. To enable automatic and interactive temperature prediction.
 - g. To derive useful insights from weather patterns.
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2. Problem Statement

Weather conditions fluctuate daily due to environmental and atmospheric changes. Raw weather data alone does not provide meaningful insights. Users also face difficulty predicting future temperature values without scientific tools.

The problem addressed in this project:

How can weather data be processed, analyzed, visualized, and used to predict future temperature in an interactive and user-friendly manner?

3. Functional & Non-Functional Requirements

Functional Requirements

1. **Load Data** – The system must read weather data from a CSV file.
2. **Data Cleaning** – Remove missing values and sort data based on date.
3. **Anomaly Detection** – Identify outliers in temperature, rainfall, and pollution.
4. **Statistical Summary** – Display descriptive statistics of the dataset.
5. **Visualization** – Plot temperature graphs, boxplots, and heatmaps.
6. **Model Training** – Train a Linear Regression model for temperature prediction.
7. **Automated Prediction** – Predict the next day's temperature.
8. **Interactive Prediction** – Allow users to enter their own parameters for prediction.
9. **Menu System** – Provide a user menu with options (1–8).

Non-Functional Requirements

1. **Usability** – Simple command-line interface.
2. **Performance** – Predictions should be computed efficiently.
3. **Maintainability** – Code should be modular and easy to modify.
4. **Reliability** – Accurate and consistent predictions.
5. **Portability** – Runs on any Python-supported environment.

4. Workflow / Flowchart

Workflow Steps

- a. Load the CSV dataset.
- b. Clean and preprocess data.
- c. Add moving average column.
- d. Display insights or statistics (if chosen).
- e. Detect anomalies.
- f. Visualize weather parameters.
- g. Train Linear Regression model.
- h. Predict future temperature.
- i. Interactive prediction via user input.